

H.W

Tracing of quick sort of 54, 12, 19, 3, 8, 46, 31

qs(a, 0, 6)

first=0, last=6, pivot=0

54	12	19	3	8	46	31
X	X	X	X	X	X	j
						i

i	j	a[i]	a[j]	a[pivot]	
0	6	54	31	54	54 <= 54
1	6	12	31	54	12 <= 54
2	6	19	31	54	19 <= 54
3	6	3	31	54	3 <= 54
4	6	8	31	54	8 <= 54
5	6	46	31	54	46 <= 54

swap(a[pivot], a[j])

so

3	12	19	8	31	46	54
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3	12	19	8
X	i	j	X

j

3	12	9	8
	j	j	

j

31

12	6	19
i	X	X
	j	i

31

46

54

i > j => swap pivot & j

3 9 12 19 31 46 54

3 9 12 19 31 46 54

i j a[i] a[j] a[pivot]

0 5 31 46 31

initial

3 4 3 8 31

p < j $\Rightarrow$ swap

0 5 31 46 31

5 4 46 3 31

p > j $\Rightarrow$ swap pivot

0 3 3 8 3

initial

1 0 12 3 3

p > j $\Rightarrow$ swap j & pivot

1 3 12 8 12

initial

2 3 19 8 12

swap p & j

1 3 12 19 12

initial

3 2 19 8 12

p > j $\Rightarrow$ swap pivot

$\therefore$ final array =

3 9 12 19 31 46 54

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